Physiological Correlates of Maternal Separation in Surrogate-Reared Infants: A Study in Altered Attachment Bonds

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Three pigtail infant monkeys (*Macaca nemestrina*) were separated from their mothers shortly after birth and raised in social isolation on cloth surrogates. At about 24 weeks of age they were surgically implanted with multichannel biotelemetry systems. Following the collection of baseline behavioral and physiological data, the cloth surrogates were removed for 4 days, then returned. The behavioral and physiological (heart rate, body temperature, sleep patterns) reaction to separation from the surrogate was much less intense than is the case in group-living pigtail infants that are separated from their mothers, suggesting that the attachment bond to a cloth surrogate may well be different than the bond to a living mother in a social group. Heart rate mean values and variability were similar in surrogate-reared and mother-reared infants, suggesting a degree of environmental independence in heart rate regulation.

The premature loss of the mother can be a major crisis for the young of many higher animals. Spitz (1946) described the syndrome of "anaclitic depression" in human infants who have been deprived of proper mothering. Goodall (1971a, 1971b) described severe behavioral disturbances, including depressive behavioral reactions, in young chimpanzees that had lost their mothers, and implicated psychological mechanisms as important causative agents. Infant monkeys of several species, when separated from their mothers, frequently exhibit an "agitation-depression" or "protest-despair" type reaction (Hinde & Spencer-Booth, 1971; Kaufman & Rosenblum, 1967a,b, 1969; Schlottman & Seay, 1972; and Seay & Harlow, 1965).

We have studied the physiological accompaniments of the agitation-depression

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reaction in group-living infant pigtail monkeys that were separated from their mothers (Reite & Short, in press, a; Reite, Short, Kaufman, Stynes, & Pauley, in press c). We found the initial agitation reaction to be accompanied by increases in both heart rate (HR) and body temperature (BT), compatible with a state of organismic hyperarousal. Sleep patterns the 1st separation night were characterized by increased sleep latency, more awake time, less total sleep time, and more frequent arousals. Although non-REM (slow wave) sleep was not appreciably affected in terms of absolute amount, rapid eye movement (REM) sleep was severely disturbed, with greatly increased REM latency, and much less (or no) REM sleep time. Most often the 1st separation night was accompanied by marked decreases in both HR and BT. The depressive behavioral reaction following separation was generally accompanied by continued decreases in both BT and HR and persistent alterations in nocturnal sleep patterns, especially REM sleep disturbances. Most of the physiological changes accompanying the agitation-depression reaction tended to return to normal following reunion with the mother, although some infants evidenced persistent separation-induced decreases in HR.

Previous studies have shown that rhesus monkey infants appear to form strong attachment bonds to cloth covered surrogate mothers (Harlow & Zimmerman, 1959). We might expect the infant-mother attachment bond (and possibly the infant) to be quite different under these conditions, however, and we might reasonably expect the physiological and behavioral reaction to separation to be quite different in the surrogate-reared infant. We also realized that normative physiological data collected from surrogate-reared infants during the baseline (pre-separation) period are of considerable interest in their own right in that they may provide further support for the relative environmental independence of the development of several physiological systems.

Methods

The subjects were 3 pigtail infants (*Macaca nemestrina;* 2 males, 1 female), born of feral-reared mothers that had all been living in the Department of Psychiatry Primate Laboratory for several years. The infants were removed from their mothers approximately 24 hr following their birth, and were housed initially in cardboard boxes lined with towels. Supplemental heat was provided by an electric heating pad in the box. They were fed from standard nursing bottles by hand. Initially, feedings were essentially on demand; as soon as the infants were able to adequately support themselves, a nursing bottle on a ringstand was added to the box and they were allowed to feed as often as they desired. They were provided with a surrogate within the 1st 2 weeks of postnatal life and evidenced behavioral attachment to this surrogate quite early. The surrogate consisted of several rolled-up cotton bath towels with a brown cloth cover.

Between the ages of 3 and 5 weeks, they were moved to individual 50 x 50 x 50-cm epoxy-covered wooden cages that were kept in a well-ventilated room where the ambient temperature was maintained at a nominal 25°C. Their cages contained only the surrogate (tied in one corner) and a plastic chain suspended from the top of the cage. No supplemental heating was provided after the move to the surrogate cage. Light was provided by a 60-W incandescent bulb mounted above 15-cm diameter,

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Plexiglas-covered holes in the top of the cage. Timers automatically turned the lights off between 2000 and 0700 hours. The surrogate cages were located in a small room that contained only 2 cages at any one time; each infant could hear sounds produced by the other, but neither could see the other, or any other animal. Thus, they were socially isolated from other monkeys, but other sensory input was only slightly attenuated. The infants were raised with a minimum of human contact, except that necessary for cage cleaning and maintenance. Cages were cleaned weekly, at which time the infant was removed, with its surrogate, and placed in a wastebasket lined with towels while the cage was cleaned in another part of the laboratory. The surrogates were changed every 6 weeks. With 2 of the infants, the existing surrogate was merely replaced with a clean one. The 3rd infant protested such a change quite vigorously, and so for that animal the clean surrogate was placed in the cage next to the old surrogate for a 24-hr period, at which time the old surrogate was removed, with the clean one remaining. By the time the infants were old enough to participate in the separation experiments described below, their feeding schedule consisted of 1 bottle of Similac (Ross Laboratories, Columbus, Ohio) and monkey biscuits at 0900 hours, and another feeding of monkey biscuits at 1600 hours.

The 3 infants were surgically implanted with multichannel biotelemetry systems of our design at the ages of 19, 24, and 28 weeks (Pauley, Reite, & Walker, 1974; Reite & Pauley, 1976). The telemetry systems transmitted eye movements (EOG), muscle activity (EMG), electrocardiogram (EKG), body temperature (BT), and 3 channels of electroencephalographic (EEG) data. The telemetered signal was received on a modified commercial FM receiver, demodulated, and the various signals recorded on paper and magnetic tape. (Our physiological data collection system has been described in detail in Reite, Pauley, Walker, Kaufman, & Stynes, 1974b.) The infants were returned to their cages following recovery from surgical anesthesia. Sutures were removed 10 days after surgery with physiological and behavioral data collection beginning shortly thereafter (see Fig. 1).

The all-night paper polygraph sleep record was hand-scored on an ASR33 teletype keyboard with the punched paper tape so generated being computer-scored (Walker & Reite, 1976), providing quantified measurements of a variety of sleep variables including total sleep time (TST), time spent awake during the night, time spent in the various sleep stages (Drowsy, Stage 2, Stage 3+4, and REM), the number of arousals during the night, sleep latency (from lights out), REM latency (from sleep onset), inter-REM interval (IRI), and the numer of REM periods (NREMPS).

Heart rate and body temperature values were obtained every 30 min during the day and every 15 min at night by the technician in attendance. A mean value for both body temperature and heart rate was obtained for each day (between 1000 and 1600 hours) and for each night (between 2200 and 0400 hours) by averaging all the discrete values obtained during these periods.

Six 5-min samples of behavior were obtained each day. A trained observer dictated ongoing behavior onto an audio cassette recorder for a 5-min period. The audio tape was later replayed with the observer listening to it through earphones, and each behavior as it occurred was entered into a PDP-12 computer (Digital Equipment Corporation) by striking a specified key on the computer keyboard. The computer then provided a statistical summary of the observation session with duration and frequency measurements of all observed behaviors (Walker & Reite, 1975).

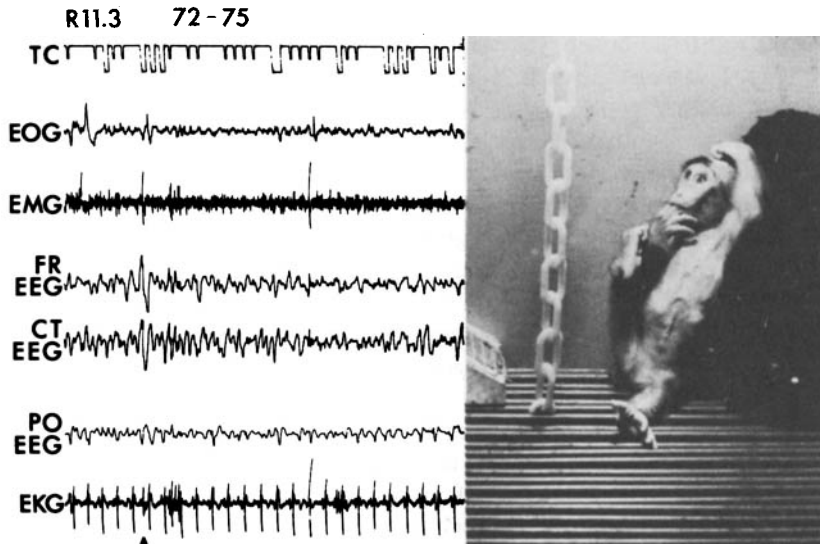


Fig. 1. A surrogate-reared pigtail infant resting on the surrogate with simultaneously recorded physiological data. The infant's posture, although somewhat unusual in appearance, is not atypical. TC = time code, EOG = eye movement, EMG = muscle activity, FR EEG = frontal EEG, CT EEG = central-temporal EEG, PO EEG = parieto-occipital EEG, EKG = electrocardiogram. The photograph was taken at the time indicated by the arrow beneath the EKG channel.

Following 4 consecutive 24-hr periods of baseline physiological and behavioral data collection, the cloth surrogate mothers were removed from the cages and the infants were left alone in the cage. The ages of the 3 infants at separation were 22, 27, and 31 weeks. Separation continued for a 4-day period at the end of which time the surrogate was returned to the cage. Data collection continued for another 3 days following reunion with the surrogate.

So that we could more easily compare the results from the surrogate-reared infants to data obtained from group-living infants (Reite et al., in press, c), we scored a comparable group of behaviors, which included motion (locomotion), movement, rest, play (with self, the chain, or other inanimate objects), and contact with the surrogate (both duration and frequency). In addition, scores were obtained for inanimate object exploration (manual), oral object exploration, and self-orality.

For each behavior, a percent of the total time observed was computed for each day. The by-day scores were then averaged across baseline days for each animal to obtain a baseline mean score. To determine if behavior changed across experimental treatments (separation and reunion), we subtracted the baseline mean from the treatment score for each animal. The subsequent scores, now in terms of a change from the normal baseline score, were averaged across animals to get an indication of those days in which behavior changed as a result of experimental manipulation.

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Results

Before we consider the effects of separation from the surrogate on the physiology and behavior of the isolation-reared infant, we will describe briefly their normal physiological patterns as recorded during the baseline period, as few physiological data are available on unrestrained isolation-reared monkeys.

Both BT and HR exhibited normal circadian variations as we have previously described in group-living infants (Reite, Pauley, Kaufman, Stynes, & Marker, 1974a). Mean BT values were similar to those found in group-living mother-reared infants. Heart rate values for the 3 isolation-reared infants fell within the range found for mother-reared group-living infants. This is not to imply that the 3 infants had the same or similar HR values, for as in group-living infants, individual variability was the rule. (These data are illustrated in Fig. 2, where the mean nocturnal HR values for the 3 surrogate-reared infants are compared to similarly derived values for 16 mother-reared group-living infants.) Individual differences were prominent in both groups, and the surrogate-reared infants, as a group, were not separable from the mother-reared infants. Daytime HR values showed a similar pattern. Daytime and nighttime HR variability measures were also quite similar in the surrogate-reared and the mother-reared group-living infants. In addition, baseline nocturnal sleep patterns in the isolation-reared infants were almost identical to sleep patterns previously reported in group-living infants (Reite & Short, in press, b; Reite, Stynes, Vaughn, Pauley, & Short, 1976). Given just the physiological measurements we obtained, we could not distinguish readily a surrogate-reared from a mother-reared group-living infant.

In general, as with group-living infants, following separation from the surrogate (which normally occurred midday) the infants exhibited an increase in motion and a decrease in rest behavior for the remainder of the 1st separation day, similar to that described in rhesus infants separated from surrogate mothers (Mason & Berkson, 1975). Notable in the surrogate-reared infants was a marked increase in the intensity of self-mouthing behavior following separation. They would literally throw their arms, hands, and feet into their mouths with dramatic intensity, appearing to all intents and

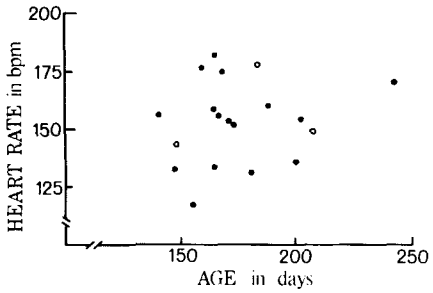


Fig. 2. Mean nocturnal (2200-0400 hours) HR values for 3 surrogate-reared (3 open circles) and 16 mother-reared group-living (black circles) pigtail infants. Each HR value represents the mean of 3 or 4 normal baseline nights. The ages of the infants are on the abcissa.

purposes as though they were biting themselves, yet they did not sustain injury. Vigorous self-mouthing behavior was not often seen in group-living infants following separation.

During the 2nd and subsequent days of separation, motion and rest behavior scores were not appreciably different from baseline; they revealed no pronounced decrease in motion as is most often seen in group-living infants during the depressive behavioral reaction that follows maternal separation. They did show, however, a slight decrease in play behavior. At no time did we observe the characteristic "slouched" posture of depression.

When the surrogate-reared infants were reunited with their cloth mothers, the frequency of contact with the surrogate increased during the 1st day, although the total time in contact with the surrogate was not greater than during baseline. Object exploration and oral object exploration decreased during the 1st day of reunion, findings consistent with those of Harlow and Zimmerman (1959) in their studies of cloth surrogate-reared rhesus infants.

Physiologically, the most prominent changes found during separation were in sleep patterns (Table 1): sleep latency and awake time increased. Total sleep time (TST), REM time, and NREMPS decreased. Few of the changes were as pronounced as those previously described in group-living infants, and with the exception of time awake and number of arousals, all had returned to within 1 standard deviation of normal baseline values by the 4th night of separation.

During the actual experience of being physically separated from their surrogates, the infants protested vigorously and vocally and showed increases in their heart rates for a short period (< 10 min) following the struggle, but they offered no evidence of a persistent increase in HR during the remainder of the 1st separation day (S_1). Body temperature did not increase during the actual separation experience, but mean BT values were slightly increased ($\approx .3°C$) during the remainder of Day S_1.

Mean daytime BT values during the remainder of separation and during reunion did not differ from baseline values. Nocturnal mean BT values, if anything, were very slightly increased during separation.

No significant changes were recorded in either daytime or nighttime mean HR values during the remainder of the separation or reunion period: mean values observed were within the baseline range of variability.

Discussion

Prior to our consideration of the effects of separation, a comment is in order regarding the basal HR values of these 3 infants. The infants' mean HR values were distributed well within the range exhibited by normal mother-reared group-living infants; the surrogate-reared infants were not separable as a group as being unusually high, low, or variable. The infants' basal heart rates also differed one from another to the same degree that group-living infants' differ one from another. These observations are most interesting, for (as we have previously shown in the case of sleep patterns in Reite & Short, in press, b) they suggest that a major environmental difference, one that is capable of leading to marked peculiarities in behavioral development, does not appear to influence the infant's basic organization and control of HR, which may, therefore, be largely intrinsically determined. These observations suggest that in certain

TABLE 1. Mean Sleep Values for 3 Surrogate-Reared Infants during the Baseline Nights Grouped ($B\bar{x}$), the 4 Consecutive Nights Following Surrogate Separation (S_1 to S_4), and 3 Consecutive Nights Following Reunion with the Cloth Surrogate (R_1 to R_3).

	$B\bar{x}$		S_1		S_2		S_3		S_4		R_1		R_2		R_3	
Number of nights	12	(30)	3	(10)	3	(10)	3	(10)	3	(10)	3	(8)	3	(8)	2	(7)
Sleep latency	16 ± 11	(35)	29	(77)	19	(37)	8	(31)	10	(39)	9	(22)	8	(26)	4	(42)
TST	590 ± 32	(563)	503	(471)	557	(507)	573	(524)	558	(536)	587	(582)	582	(578)	598	(562)
Awake	47 ± 19	(51)	119	(95)	70	(94)	72	(85)	90	(66)	41	(42)	60	(46)	55	(45)
Number of arousals	23 ± 8	(15)	43	(33)	38	(27)	31	(21)	34	(21)	37	(14)	20	(15)	24	(16)
Drowsy	40 ± 13	(20)	44	(27)	41	(22)	49	(27)	40	(26)	37	(26)	40	(33)	43	(29)
Stage 2	303 ± 40	(297)	283	(269)	287	(295)	306	(270)	286	(294)	317	(297)	304	(286)	286	(299)
Stage 3-4	152 ± 22	(151)	138	(146)	167	(146)	138	(169)	145	(162)	143	(164)	143	(159)	161	(141)
SWS	495 ± 34	(469)	465	(442)	495	(463)	493	(466)	471	(484)	497	(487)	487	(478)	490	(469)
REM	91 ± 14	(90)	36	(21)	60	(36)	78	(48)	84	(46)	89	(88)	93	(95)	108	(89)
REM latency	68 ± 21	(64)	219	(212)	108	(163)	180	(197)	83	(119)	49	(102)	79	(80)	34	(77)
IRI	65 ± 6	(58)	61	(111)	75	(107)	69	(83)	62	(93)	62	(60)	74	(55)	64	(60)
NREMPS	9.2 ± .8	(9.7)	5.7	(3.6)	7.7	(5.1)	7.7	(6.2)	9.3	(7.2)	10.7	(8.4)	8.7	(10.6)	10.5	(9.9)

Baseline mean values are followed by the standard deviation. All values in minutes except number of nights (number of nights recorded), number of arousals (number of arousals during the night), and NREMPS (number of REM periods). SWS (slow wave sleep) comprises drowsy, Stage 2, and Stage 3-4 sleep. All values do not add up exactly due to computer round-off error. The values in parentheses are those we have previously found in group-living infants that have been separated from their mothers (Reite & Short, in press, a), and are presented here for comparative purposes.

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respects the animal's neurobiological ontogenetic trajectory (see Waddington, 1968) may have progressed fairly normally in spite of the atypical environment and the appearance of the disordered behavior.

Our observations following separation from the surrogate, both behavioral and physiological, are generally in marked contrast to previous studies of the effects of maternal separation in mother-reared group-living pigtail infants. Although behavioral and physiological arousal was evident, possibly an attenuated "agitation" reaction immediately following separation, neither behavioral or physiological accompaniments of a subsequent depressive reaction were observed.

The changes in sleep patterns were similar in form and direction to the changes found in group-living infants following maternal separation, but they were generally of lesser magnitude and of shorter duration. The sleep pattern changes were in fact quite similar to those described (in man) as constituting the "first night effect," which result from a change in sleep environment (Agnew, Webb, & Williams, 1966). We might fairly conclude that such an effect could have accounted for the changes found in these 3 infants.

Thus, one might conclude that the infant-surrogate attachment bond, though seemingly both real and strong to the observer of the dyad, is not of sufficient intensity to produce, when disrupted, a particular marked psychobiological reaction.

The reaction to separation is a complex multidimensional phenomenon, however, likely involving, in addition to the attachment bond per se, individual differences within the organism (including possible species differences) and differences attributable to the specific environment in which the separation is carried out or in which the separated infant finds itself. Chappell and Meier (1975) have shown that the nature of the behavioral response to maternal separation in rhesus infants is significantly influenced by the housing conditions during the separation. In a similar fashion, in group-living bonnet monkeys (*M. radiata*) the nature of the social group in which the separated infant finds itself may be a most important determinant of the behavioral reaction following separation (Kaufman & Stynes, 1977).

Our data do indicate that from the standpoint of the behavioral and physiological reaction to separation, the loss of the surrogate by an isolation-reared infant is by no means as significant an event as is the loss of the living mother by a group-living pigtail infant. Although this may be due in part to an alteration in the nature and/or intensity of the attachment bond, the contribution of other influences, including those mentioned above, will have to be determined. As such data become available, and as we begin to complete the puzzle, we will approach an understanding of the role of attachment bonds in the developing organism.

Notes

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References

Agnew, H. W., Webb, W. B., and Williams, R. L. (1966). The first night effect: An EEG study. *Psychophysiol., 2*: 263-266.

Chappell, P. F., and Meier, G. W. (1975). Modification of the infant response to separation through manipulation of the physical environment. *Biol. Psychiat., 10*: 644-657.

Goodall, J. v.-L. (1971a). Some aspects of mother-infant relationships in a group of wild chimpanzees. In H. R. Schaffer (Ed.), *The Origins of Human Social Relations.* New York: Academic Press. Pp. 115-128.

Goodall, J. v.-L. (1971b). *In the Shadow of Man.* Boston: Houghton Mifflin Co.

Harlow, H. F., and Zimmerman, R. R. (1959). Affectional responses in the infant monkey. *Science, 130*: 421-432.

Hinde, R. A., and Spencer-Booth, Y. (1971). Effects of brief separation from mother on rhesus monkeys. *Science, 173*: 111-118.

Kaufman, I. C., and Rosenblum, L. A. (1967a). Depression in infant monkeys separated from their mothers. *Science, 155*: 1030-1031.

Kaufman, I. C., and Rosenblum, L. A. (1967b). The reaction to separation in infant monkeys: Anaclitic depression and conservation-withdrawal. *Psychosom. Med., 29*: 649-675.

Kaufman, I. C., and Rosenblum, L. A. (1969). Effects of separation from mother on the emotional behavior of infant monkeys. *Ann. N.Y. Acad. Sci., 159*: 681-695.

Kaufman, I. C., and Stynes, A. J. (1977). Depression induced in a bonnet macaque infant by separation. Presented at the Annual Meeting of the American Association for the Advancement of Science, Denver, Colorado, Feburary.

Mason, W. A., and Berkson, G. (1975). Effects of maternal mobility on the development of rocking and other behaviors in rhesus monkeys: A study with artificial mothers. *Dev. Psychobiol., 8*: 197-211.

Pauley, J. D., Reite, M. L., and Walker, S. D. (1974). An implantable multi-channel biotelemetry system. *Electroencephalogr. Clin. Neurophysiol., 37*: 153-160.

Reite, M., and Pauley, J. D. (1976). Multichannel implantable biotelemetry: Problems, pitfalls, and rewards. In T. B. Fryer, H. A. Miller, and H. Sandler (Eds.), *Biotelemetry III.* New York: Academic Press. Pp. 83-86.

Reite, M., Pauley, J. D., Kaufman, I.C., Stynes, A. J., and Marker, V. (1974a). Normal physiological patterns and physiological behavioral correlations in unrestrained monkey infants. *Physiol. Behav., 12*: 1021-1033.

Reite, M., Pauley, J. D., Kaufman, I. C., Stynes, A. J., and Marker, V. (1974b). A systems approach to studying physiology and behavior in monkey infants. *J. App. Physiol., 37*: 417-423.

Reite, M., Stynes, A. J., Vaughn, L., Pauley, J. D., and Short, R. (1976). Sleep in infant monkeys: Normal values and behavioral correlates. *Physiol. Behav., 16*: 245-251.

Reite, M., and Short, R. (in press, a). Nocturnal sleep in separated monkey infants. *Arch. Gen. Psychiat.*

Reite, M., and Short, R. (in press, b). Nocturnal sleep in isolation reared monkeys: Evidence for environmental independence. *Dev. Psychobiol.*

Reite, M., Short, R., Kaufman, I. C., Stynes, A. J., and Pauley, J. D. (in press c). Heart rate and body temperature in separated monkey infants. *Biol. Psychiat.*

Schlottman, R., and Seay, B. (1972). Mother infant separation in the Java Monkey (*Macaca irus*). *J. Comp. Physiol. Psychol., 79:* 334-340.

Seay, B., and Harlow, H. (1965). Maternal separation in the rhesus monkey. *J. Nerv. Ment. Dis., 140*: 434-441.

Spitz, R. A. (1946). Anaclitic depression. *Psychoanal. Study Chile, 2*: 313.

Waddington, C. H. (1968). The basic ideas of biology. In C. H. Waddington (Ed.), *Toward a Theoretical Biology I. Prolegomena.* Edinburgh, Scotland: Edinburgh University Press. P. 12.

Walker, S., and Reite, M. (1975). Computer correlation of telemetered physiological data with behavior. *Biomed. Sci. Instrum., 11*: 175-178.

Walker, S., and Reite, M. (1976). SLEEPSTA: Sleep stage data handler. *Psychophysiology, 13*: 159.

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